[Inch-Pound]
A-A-55859
JULY 1,1996

# COMMERCIAL ITEM DESCRIPTION

## TUBE, NONMETALLIC; POLYVINYL CHLORIDE (PVC) FLEXIBLE (GENERAL USE)

The General Services Administration has authorized the use of this Commercial Item Description (CID) by all federal agencies.

1. SCOPE. This CID covers clear flexible PVC tubing intended for use in general applications.

2. CLASSIFICATION.

2.1 Types.

Type 1 For use with petroleum products.

Type 2 For use with food and consumable products.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., or any data which may improve this document should be sent to: Commander, Defense Supply Center, Columbus, ATTN DSCC-VSS, P.O. Box 3990, Columbus, OH 43216-5000.

AMSC N/A FSC 4720

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2.1.1 Size and wall thickness. Plastic tubing shall be specified as described below:

a. Nominal Inside Diameter (ID) shall be specified in 1/64 of 1 inch.
Example: 5 = 5/64 of one inch (0.0780).

b. Nominal Wall Thickness (WT) shall be specified in 1/64 of 1 inch.
Example: 10 = 10/64 of one inch (0.1560)

c. Metric sizes shall be designated by preceding the ID and WT designators with an (M). Metric sizes shall be rounded to one decimal place (Example M125 = 12.5 mm)

d. Tolerance on inside diameter for all sizes shall be ±.0156 inch (0.4 mm).

e. Tolerance for wall thickness for all sizes shall be minus 0 or plus .0156 inch (0.4 mm).

3. SALIENT CHARACTERISTICS.

3.1 General. Tubing provided in accordance with this CID shall be colorless, clear, flexible Pvt.

3.2 Lengths. Tubing shall be furnished in the length specified (see Table 1 and paragraph 7.1c).

TABLE 1. Tubing lengths.

| Length Code | Length |
|---|---|
| A | 50 Ft (15,240 mm) |
| B | 100 Ft (30,480 mm) |
| c | 250 Ft (76,200 mm) |
| D | 500 Ft (152,400 mm) |
| E | 1000 Ft (304,800 mm) |

3.3 Form. Tubes shall be furnished in the form specified (see Table 2 and paragraph 7.lc).

TABLE 2. Tubing forms.

| Form Code | Form |
|---|---|
| 1 | Coils |
| 2 | Reels |



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3.4 Flexibility. Tubing shall be capable of bending around a mandrel equal to five times the Outside Diameter (OD) of the tube without collapsing or restricting flow.

3.5 Physical properties. Physical properties of the tubes shall be in accordance with Table 3.

TABLE 3. Physical properties.

| PROPERTY | REQUIREMENT | TEST METHOD (ASTM) |
|---|---|---|
| Brittleness temperature, °C | -46°C | D746 |
| Specific gravity | 1.15-1.80. | D792 |
| Tensile strength, p.s.i. | 1800-2100 | D638 |
| Elongation, percent | 300 min. | D638 |

3.6 Chemical resistance. Tubing shall not discolor, flake, crack or pit when tested in accordance with USP-NF Physicochemical Tests-Plastics, ASTM D1003, Standard Method of Test for Haze and Luminous Transmittance of Transparent Plastics, and ASTM E308, Tentative Method of Test for Luminous Reflectance, Transmittance, and Color of Materials.

3.7 Heat resistance. Tubing shall be capable of withstanding temperatures of 93±3 degrees Centigrade for one hour without visual signs of deterioration such as discoloring, flaking, cracking, pitting, hardening, or melting.

3.8 Metal content. The use of Hazardous metal such as lead shall be eliminated or minimized where possible.

3.9 Workmanship. Tubes shall be free from blemishes and flaws which mar their appearance or affect their serviceability.

4 REGULATORY REQUIREMENTS.

4.1 Relations and Laws. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).



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5. QUALITY ASSURANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market Acceptability. The Contractor/Supplier must have provided this item as his commercial product for a period not less than one year.

6. PACKAGING.

6.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with ASTM D3951 or as specified in the contractor order (see 7.1d).

6.2 Marking and labeling. Marking and labeling shall be commercial or as specified in the contract or order (see 7.1e).

7. NOTES.

7.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this CID.
b. Inside Diameter (ID) and Wall Thickness (WT).
c. Length and form required (see 3.1 and 3.3).
d. Applicable level of preservation, packaging, and packing (see 6.1).
e. Applicable marking and labeling (see 6.2).
f. Quantity of tubing required.

7.2 Part Identification Number (PIN). The following CID based PIN procedure is for Government purposes and does not constitute a requirement for contractors/suppliers. The PIN shall be constructed as follows:



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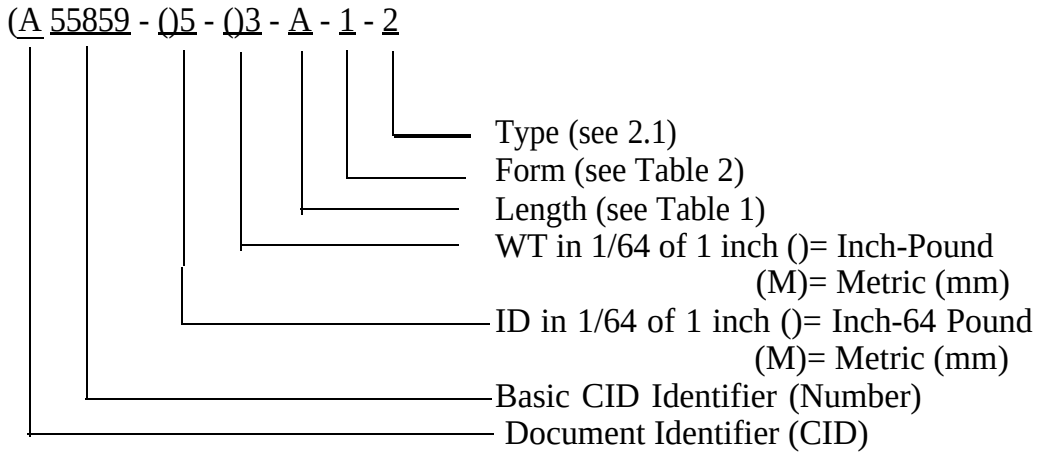


(Example: A55859-5-3-A-1-2) = Clear, flexible PVC tube, 5/64 inch nominal ID and 3/ inch (0.0468) nominal WT, furnished in a 50 foot length coil, food and consumable compatible.)

(Example: A55859-M100-M50-A-1-2 = Clear, flexible PVC tube, Metric ID of 10.0 mm and a Metric WT of 5.0 mm, furnished in 15,240 mm length coil, food and consumable compatible.)

7.3 ASTM documents. ASTM documents cited may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.

7.4 Other related documents. For information only, other related documents (not requirements of this specification) are: A-A-55697 (PVC Tubing for Laboratory and Medical use), International Standards Organization (ISO) 1135-4, for transfusion sets, and ISO 8536-4 for infusions sets.

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